Carrier Origin and Steganographic Detectability

Does the source of an image change how easily we can detect hidden data?

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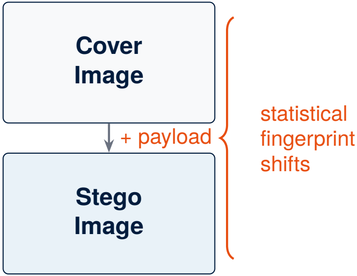
The Big Picture

The intuition

Hiding data in an image shifts its statistical fingerprint — even by a tiny amount, detectors can pick it up.

The twist

What if that fingerprint was **already different** before any data was embedded?
AI-generated images and real photographs may not look the same to a detector, even in their unmodified state.



Core question: Does the *origin* of the carrier — real photo vs. AI-generated — change how easily we detect hidden data?

Research Questions

	Question	Type	Description
RQ1	Carrier Source	Primary	Does carrier source (real vs. ML-generated) affect detectability under matched embedding?
RQ2	Generator Effect	Primary	Within ML carriers, does the choice of generator (SDXL vs. FLUX.1) matter?
RQ3	Payload Interaction	Exploratory	Does increasing payload size widen or close the detectability gap?
RQ4	Embedding Branch	Exploratory	Do spatial (LSB + PNG) and frequency (DCT + JPEG) branches show different effects?
RQ5	Encryption Invariance	Verification	Does AES-256-CBC encryption of the payload affect detectability?

These five questions structure the entire experiment and map directly to pipeline outputs.

Prototype

Vertical Prototype — Depth

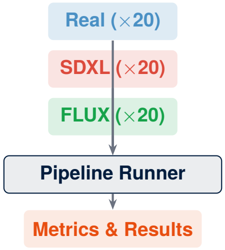
Each core component implemented and tested in isolation:

- **LSB embedding** — configurable fill rate and bit depth
- **AES-256-CBC encryption** — payload-level crypto
- **3 statistical detectors:** Chi-Square, RS Analysis, Sample Pairs

Proves each component works independently.

Horizontal Prototype — Breadth

All components wired into one pipeline running across **60 images**:



Proves all parts work together; early results across all five RQs.

Code Architecture

`data/` Acquisition	COCO + diffusers
`embedding/` LSB + DCT	bit masking + AES
`detection/` Detectors	χ^2, RS, SP
`evaluation/` Metrics	AUC, EER, breakdowns
`pipeline/` CLI Runner	single command

Launching the viewer

```
$ pip install -r requirements.txt
$ python viewer.py
Serving Stego Explorer on
http://localhost:8765

Press Ctrl+C to stop.
```

What the viewer provides

- Runs overview with source-effect delta
- Per-run detail: config, results, gallery, conditions
- RQ-organized analysis cards with AUC bars
- Per-image detector scores and cover/stego comparison
- Context-aware search across all pages